

Two-Stage Binary Correction for Nearly Linear Lattice Coverings

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Abstract

Our companion paper proves that every n -dimensional convex body K satisfies $\theta_L(K) \leq Cn \log n (\log \log n)^{10/3+o(1)}$. We refine its vertical construction by replacing the one-step high-rank finite-field correction with a two-stage finite-group scheme and obtain

$$\theta_L(K) \leq Cn \log n (\log \log n)^{\frac{2}{3} \log_2(2\pi e) + o(1)}, \quad \frac{2}{3} \log_2(2\pi e) = 2.72946 \dots$$

At constant load, a random linear subspace first sends a positive proportion of the finite grid into a folded-Gaussian information layer. An exponential-potential argument then chooses $O(\log |G|)$ binary translations so that every element of the ambient finite abelian group has a positive proportion of good labels. We encode these translations by $O(\log d)$ orthogonal label coordinates and use Hoeffding's inequality to select a label with controlled Gaussian weight. This gives the uniform normalized vertical weight $d^{-\frac{1}{2} \log_2(2\pi e) + o(1)}$ while preserving the normalized Boolean law and its entropy estimates. Because the Boolean directions have rank d whereas the enlarged vertical lattice has dimension $d + O(\log d)$, we also prove the required partial-rank weighted lifting theorem. The finite-group, label-coordinate, and lifting arguments are self-contained; the analytic and horizontal ingredients are cited from the companion paper in the precise forms used here.

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1 Introduction

A *convex body* is a compact convex set with nonempty interior. For a full-rank lattice $\Lambda \subseteq \mathbb{R}^n$, write $\det \Lambda$ for its positive covolume. The *lattice covering density* of a convex body $K \subseteq \mathbb{R}^n$ is

$$\theta_L(K) := \inf \left\{ \frac{\text{vol}_n(K)}{\det \Lambda} : \Lambda \subseteq \mathbb{R}^n \text{ is a lattice and } \Lambda + K = \mathbb{R}^n \right\}.$$

All logarithms are natural unless a base is displayed, and all asymptotic statements involving only the auxiliary dimension d are taken as $d \rightarrow \infty$.

In the companion paper [5], we prove the universal estimate

$$\theta_L(K) \leq Cn \log n (\log \log n)^{10/3+o(1)}. \quad (1.1)$$

That proof separates a low-dimensional vertical space from a high-dimensional horizontal space. In the vertical space, a folded Gaussian assigns weights to the vertices of a Boolean cube, and a movable information layer supplies centers with controlled total mass and entropy. A high-rank finite-field Kakeya theorem of Kopparty, Lev, Saraf, and Sudan [4] moves every finite-field grid cell directly into this layer. The resulting normalized Gaussian weight above every vertical translate is $d^{-5/2+o(1)}$. The horizontal argument converts these weights into volumes of sections, finds one horizontal lattice with small holes for every relevant section label, amplifies the hole estimates along the Boolean cube, inserts the horizontal shifts into a shear lattice, and applies Rogers' completion lemma [9].

The exponent $10/3$ comes from two scales. The min-entropy condition forces

$$d = (\log \log n)^{4/3+o(1)}, \quad (1.2)$$

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while the reciprocal of the vertical weight contributes $d^{5/2+o(1)}$. More generally, a uniform vertical weight $d^{-a+o(1)}$, together with the same entropy and geometric estimates, leads to an exponent $4a/3$ on $\log \log n$.

The purpose of this note is to replace the direct finite-field cover by two corrections with different roles.

- (1) At constant average load, a random linear subspace moves a fixed positive proportion of grid cells into the information layer.
- (2) In any finite abelian group, $O(\log |G|)$ binary translations can be chosen so that every group element has a fixed positive proportion of corrections landing in a prescribed positive-density set.
- (3) The translations are realized by one orthogonal label coordinate per binary choice, increasing the vertical dimension by only $O(\log d)$.
- (4) Among the good binary labels, Hoeffding's inequality selects one whose Gaussian weight is close to the coordinatewise geometric mean. Optimizing the label spacing gives the one-coordinate factor $(\pi e)^{-1/2}$.

The first-stage folded-Gaussian weight is $d^{-1/2+o(1)}$. The number of binary labels is $d^{1+o(1)}$, so the second-stage Gaussian cost is

$$d^{-\frac{1}{2} \log_2(\pi e) + o(1)}.$$

Their product is

$$d^{-a_*+o(1)}, \quad a_* = \frac{1}{2} + \frac{1}{2} \log_2(\pi e) = \frac{1}{2} \log_2(2\pi e).$$

Substituting d from (1.2) gives the exponent

$$\alpha_* = \frac{4}{3} a_* = \frac{2}{3} \log_2(2\pi e) = 2.729460780 \dots$$

The main result of this companion note is the following.

Theorem 1.1 (Two-stage correction bound). *There are an absolute constant $C > 0$ and a nonnegative sequence $\varepsilon_n \rightarrow 0$ such that, for every sufficiently large n and every n -dimensional convex body K ,*

$$\theta_L(K) \leq Cn \log n (\log \log n)^{\alpha_* + \varepsilon_n}, \quad \alpha_* = \frac{2}{3} \log_2(2\pi e).$$

The sequence ε_n is deterministic and independent of K .

This note is not intended to duplicate the analytic part of the companion paper. We quote its folded-Gaussian layer, pointwise marginal, section-discretization, mean-hole, weighted-amplification, entropy-pruning, and completion results, and we indicate their exact role in the final assembly. The new finite-group and lattice arguments are independent of the convex body and are proved in full.

The proof is organized as follows. [Section 2](#) records the folded-Gaussian and horizontal results used from the companion paper. [Section 3](#) constructs a constant-load partial finite-field cover. [Section 4](#) proves the robust binary translation theorem. [Section 5](#) analyzes the Gaussian label coordinates. [Section 6](#) combines the two corrections into a vertical lattice. [Section 7](#) proves the partial-rank lifting theorem required because the Boolean rank is d while the vertical ambient dimension is $d + O(\log d)$. Finally, [Section 8](#) verifies every hypothesis of the horizontal argument and proves [Theorem 1.1](#).

2 Inputs from the Li–Liu construction

We first record the precise part of the folded-Gaussian construction that will be reused. These statements combine Lemmas 3.1–3.4, Proposition 3.5, and Lemma 3.7 of the companion paper [5].

For a positive integer q , let

$$\gamma_q(x) := (2\pi)^{-q/2} \exp\left(-\frac{\|x\|_2^2}{2}\right), \quad x \in \mathbb{R}^q,$$

be the standard Gaussian density on $\mathbb{R}^q$. Put $\tau := (2 \log d)^{-1}$, let

$$\gamma_\tau(u) := (2\pi\tau)^{-1/2} e^{-u^2/(2\tau)}, \quad P_\tau := \int_{-1}^1 \gamma_\tau(u) \, du,$$

and let $g(u) := P_\tau^{-1} \gamma_\tau(u) \mathbf{1}_{[-1,1]}(u)$. The folded density f and its Bernoulli parameter are

$$f(t) := g(-t) + g(1-t), \quad p(t) := \frac{g(1-t)}{f(t)}, \quad t \in [0, 1].$$

For $t = (t_1, \dots, t_d) \in [0, 1]^d$, define the product probability measure P_t on $\{0, 1\}^d$ by

$$P_t(\eta) := \prod_{i=1}^d p(t_i)^{\eta_i} (1 - p(t_i))^{1-\eta_i}.$$

For a probability vector q , write

$$H(q) := - \sum_x q(x) \log q(x), \quad H_\infty(q) := - \log \max_x q(x).$$

The differential entropy of f is $h(f) := - \int_0^1 f(t) \log f(t) dt$. Define the expected one-coordinate Shannon and min-entropies by

$$s := \int_0^1 f(t) (-p(t) \log p(t) - (1-p(t)) \log(1-p(t))) dt,$$

$$s_\infty := \int_0^1 f(t) (-\log \max\{p(t), 1-p(t)\}) dt.$$

For each sufficiently large d , fix a prime $Q = Q(d)$ satisfying

$$d(\log d)^2 \leq Q \leq 3d(\log d)^2. \quad (2.1)$$

Such a prime exists by Bertrand's postulate.

Proposition 2.1 (Folded-Gaussian layer input). *There are absolute constants $c, C, W > 0$ and quantities $s = s(d)$ and $s_\infty = s_\infty(d)$ such that*

$$h(f) = \frac{1}{2} \log \frac{\pi e}{\log d} + O\left(\frac{d^{-1/4}}{\sqrt{\log d}}\right) \quad (2.2)$$

and

$$s \asymp s_\infty \asymp \frac{d^{-1/4}}{\sqrt{\log d}}, \quad (2.3)$$

For every $r \in [-3 \log d, 3 \log d]$, there is a measurable information layer $\mathcal{G}(r) \subseteq [0, 1]^d$ with volume $v(r)$ satisfying

$$c d^{-1/2} e^{dh(f)+r} \leq v(r) \leq C d^{-1/2} e^{dh(f)+r}. \quad (2.4)$$

Every $t \in \mathcal{G}(r)$ satisfies

$$e^{-dh(f)-r-W} \leq f^{\otimes d}(t) \leq e^{-dh(f)-r}, \quad (2.5)$$

$$H_\infty(P_t) \geq ds_\infty - C \sqrt{ds_\infty \log d}, \quad (2.6)$$

$$H(P_t) \leq ds + C \sqrt{ds \log d}. \quad (2.7)$$

Moreover,

$$P_\tau^d = 1 - o(1). \quad (2.8)$$

If $t, t' \in [0, 1]^d$ satisfy $\|t - t'\|_\infty \leq Q^{-1}$, then

$$|\log f^{\otimes d}(t) - \log f^{\otimes d}(t')| \leq \Delta_d, \quad (2.9)$$

$$|H(P_t) - H(P_{t'})| + |H_\infty(P_t) - H_\infty(P_{t'})| \leq C \Delta_d, \quad (2.10)$$

where

$$\Delta_d := \frac{d}{Q^\tau} = O\left(\frac{1}{\log d}\right).$$

All error terms in this proposition are deterministic functions of d .

The following exact identity separates the total weight of the Boolean cube from its normalized vertex law.

Lemma 2.2 (Folded full-cube identity). *For every $t \in [0, 1]^d$,*

$$\sum_{\eta \in \{0,1\}^d} \gamma_\tau^{\otimes d}(t - \eta) = P_\tau^d f^{\otimes d}(t),$$

and

$$\frac{\gamma_\tau^{\otimes d}(t - \eta)}{P_\tau^d f^{\otimes d}(t)} = P_t(\eta).$$

Proof. For $e \in \{0, 1\}$ and $t \in [0, 1]$, the definitions give

$$g(e - t) = f(t)p(t)^e(1 - p(t))^{1-e}.$$

Taking products over the coordinates and summing over $\eta \in \{0, 1\}^d$ gives

$$\sum_{\eta} g^{\otimes d}(\eta - t) = f^{\otimes d}(t).$$

Since g is even and $g(u) = P_\tau^{-1} \gamma_\tau(u)$ on $[-1, 1]$, the first identity follows. Dividing the pointwise product factorization by the sum gives the second identity. $\square$

We also use the horizontal part of the companion paper in the following precise places. Its Proposition 4.1 discretizes all horizontal sections above a ball in a q -dimensional vertical space using at most $(Cm^2)^q$ labels; it is based on the pointwise marginal theorem of Eldan and Klartag [1, Theorem 1]. Its Lemma 4.2 extracts one horizontal lattice from the Rogers–Schmidt mean-hole estimate in the formulation of Ordentlich, Regev, and Weiss [6, Theorem 2.3]; the estimate originates in work of Rogers [8] and Schmidt [10]. Write $c_{\text{RS}} > 0$ for the absolute constant appearing in that lemma. Its Theorem 4.3 and Lemma 4.5 provide weighted Boolean amplification and entropy pruning, respectively. Finally, Rogers’ completion lemma [9, Lemma 4] converts the resulting almost covering into an exact lattice covering. In Section 8, we apply these results with vertical dimension $D = d + O(\log d)$ and verify their dimension, radius, entropy, load, and counting hypotheses explicitly.

3 Constant-load partial correction

We first match a movable information layer to a constant finite-field load, rather than to the high-rank Kakeya threshold.

Lemma 3.1 (Constant-load layer matching). *Fix a sufficiently large absolute constant L_0 . For all sufficiently large d , there are*

$$r \in [0, 2 \log Q], \quad k \in \{2, \dots, d - 1\}, \quad B \subseteq \mathbb{F}_Q^d$$

such that every grid cell

$$\mathcal{Q}_b := \frac{b}{Q} + [0, Q^{-1})^d, \quad b \in B,$$

meets $\mathcal{G}(r)$ and, with $\beta := |B|/Q^d$,

$$L_0 \leq \beta Q^k \leq 3L_0, \tag{3.1}$$

while

$$c v(r) \leq \beta \leq C v(r). \tag{3.2}$$

For each $b \in B$, one may fix a representative $t_b \in \mathcal{G}(r) \cap \mathcal{Q}_b$.

Proof. Put $v_* := d^{-1/2} e^{dh(f)}$. By (2.2),

$$\log v_* = - \left(\frac{1}{2} + o(1) \right) d \log \log d.$$

Consequently, $v_*Q^2/L_0 \rightarrow 0$ and $v_*Q^{d-1}/L_0 \rightarrow \infty$. Since increasing k multiplies the latter expression by Q , there is a $k \in \{2, \dots, d-1\}$ such that

$$Q^{-1} \leq \frac{v_*Q^k}{L_0} < 1.$$

Let c be the lower constant in (2.4), and fix an absolute constant $A_0 \geq \max\{2, 4/c\}$. Set

$$r := \log \frac{A_0 L_0}{v_*Q^k}.$$

The choice of k gives $\log A_0 < r \leq \log(A_0 Q)$. Thus $r \in [0, 2 \log Q]$ for large d , and (2.1) also gives $r \leq 3 \log d$, as required by Theorem 2.1. Equation (2.4) gives

$$cA_0 L_0 Q^{-k} \leq v(r) \leq CA_0 L_0 Q^{-k}.$$

Let S_0 be the set of grid cells meeting $\mathcal{G}(r)$. Since the half-open cells are disjoint and cover $[0, 1]^d$,

$$|S_0| \geq v(r)Q^d \geq 4L_0 Q^{d-k}.$$

Choose $B \subseteq S_0$ with $|B| = \lceil L_0 Q^{d-k} \rceil$. Because $k \leq d-1$, the rounding error satisfies $Q^{k-d} \leq Q^{-1}$. Hence, for large d ,

$$L_0 \leq \beta Q^k \leq L_0 + Q^{-1} \leq 3L_0.$$

Both β and $v(r)$ are within absolute constant factors of $L_0 Q^{-k}$, proving (3.2). For each $b \in B$, the definition of S_0 supplies the required representative t_b . $\square$

We now use a random k -dimensional subspace only to cover a positive proportion of the finite group.

Lemma 3.2 (Constant-load partial linear cover). *Let $B \subseteq \mathbb{F}_Q^d$ have density β , and let $1 \leq k < d$. Put*

$$\mathsf{L} := \beta Q^k.$$

Then there is a k -dimensional linear subspace $C_0 \leq \mathbb{F}_Q^d$ such that

$$\frac{|B + C_0|}{Q^d} \geq \frac{\mathsf{L}}{1 + \mathsf{L}}. \quad (3.3)$$

Proof. Choose a k -dimensional subspace C uniformly at random and choose $z \in \mathbb{F}_Q^d$ uniformly and independently. Let

$$N := |B \cap (z + C)|.$$

For every fixed C , averaging over z gives $\mathbb{E}_z N = \beta Q^k = \mathsf{L}$, and hence $\mathbb{E} N = \mathsf{L}$.

For distinct $b, b' \in B$, the event that both lie in $z + C$ requires $b - b' \in C$ and then occurs for a proportion Q^{k-d} of z . Since

$$\mathbb{P}(b - b' \in C) = \frac{Q^k - 1}{Q^d - 1} < Q^{k-d},$$

we obtain

$$\mathbb{E} N(N - 1) \leq \mathsf{L}^2.$$

Thus $\mathbb{E} N^2 \leq \mathsf{L} + \mathsf{L}^2$. The Paley–Zygmund inequality [7] gives

$$\mathbb{P}(N > 0) \geq \frac{(\mathbb{E} N)^2}{\mathbb{E} N^2} \geq \frac{\mathsf{L}}{1 + \mathsf{L}}.$$

The left side is the average, over C , of the density of $B + C$. Hence some C_0 satisfies (3.3). $\square$

Fix the objects in Theorems 3.1 and 3.2 and write

$$S := B + C_0, \quad \sigma := \frac{|S|}{Q^d}.$$

Fix L_0 once and for all large enough that

$$\sigma \geq \sigma_0 := \frac{L_0}{1 + L_0} > 0. \quad (3.4)$$

Let $\text{red}_Q: \mathbb{Z}^d \rightarrow \mathbb{F}_Q^d$ denote coordinatewise reduction modulo Q . The Construction A lattice associated with C_0 is

$$\Lambda_0 := \frac{1}{Q} \text{red}_Q^{-1}(C_0) = \mathbb{Z}^d + \left\{ \frac{\widehat{c}}{Q} : c \in C_0 \right\}, \quad (3.5)$$

where $\widehat{c} \in \{0, \dots, Q-1\}^d$ is the standard integer representative of c . This is the usual Construction A lift of a linear code [2]. Since $\Lambda_0/\mathbb{Z}^d \cong C_0$, its covolume is

$$\det \Lambda_0 = Q^{-k}. \quad (3.6)$$

4 A robust binary translation lemma

The second correction stage is purely finite-group theoretic and does not use linear structure.

Theorem 4.1 (Robust binary translation). *Let $\mathbf{G}$ be a finite abelian group with $|\mathbf{G}| \geq 2$, let $0 < \sigma_0 \leq 1$, and let $S \subseteq \mathbf{G}$ have density at least σ_0 . There are constants $C_{\sigma_0} < \infty$ and $\alpha_{\sigma_0} > 0$, depending only on σ_0 , and elements*

$$w_1, \dots, w_\ell \in \mathbf{G}$$

such that, with $M := 2^\ell$,

$$M \leq C_{\sigma_0} \log |\mathbf{G}|, \quad (4.1)$$

and, for every $x \in \mathbf{G}$,

$$\left| \left\{ \varepsilon \in \{0, 1\}^\ell : x - \sum_{i=1}^\ell \varepsilon_i w_i \in S \right\} \right| \geq \alpha_{\sigma_0} M. \quad (4.2)$$

Proof. For already chosen $w_1, \dots, w_j$, define

$$N_j(x) := \left| \left\{ \varepsilon \in \{0, 1\}^j : x - \sum_{i=1}^j \varepsilon_i w_i \in S \right\} \right|.$$

Set

$$\Phi_j := \mathbb{E}_{x \in \mathbf{G}} e^{-N_j(x)}.$$

If $\sigma := |S|/|\mathbf{G}|$, then initially

$$\Phi_0 = 1 - \sigma + \sigma e^{-1} \leq 1 - \sigma_0 + \sigma_0 e^{-1} =: \rho_0 < 1.$$

If a new translation $w \in \mathbf{G}$ is added, then

$$N_{j+1}(x) = N_j(x) + N_j(x - w).$$

Averaging over $w \in \mathbf{G}$ and using that $(x, x - w)$ is uniform on $\mathbf{G}^2$ gives

$$\mathbb{E}_w \Phi_{j+1}(w) = \left(\mathbb{E}_x e^{-N_j(x)} \right)^2 = \Phi_j^2.$$

Hence one may choose w_{j+1} so that $\Phi_{j+1} \leq \Phi_j^2$. Inductively,

$$\Phi_\ell \leq \rho_0^{2^\ell} = \rho_0^M. \quad (4.3)$$

Put $c_0 := -\log \rho_0 > 0$ and $\alpha_{\sigma_0} := c_0/4$. Choose $M = 2^\ell$ to be the smallest power of two satisfying

$$M > \frac{2 \log |\mathbf{G}|}{c_0}.$$

Its minimality gives $M \leq 4c_0^{-1} \log |\mathbf{G}|$, which proves (4.1). If (4.2) failed at some x , then

$$\Phi_\ell \geq |\mathbf{G}|^{-1} e^{-\alpha_{\sigma_0} M} > e^{-c_0 M}.$$

The strict inequality follows from $(c_0 - \alpha_{\sigma_0})M > \log |\mathbf{G}|$. This contradicts (4.3). $\square$

Apply Theorem 4.1 to $\mathbf{G} = \mathbb{F}_Q^d$ and the fixed lower density σ_0 in (3.4). The choice of M in the proof gives

$$cd \log Q \leq M = 2^\ell \leq Cd \log Q = d^{1+o(1)}, \quad \ell = (1 + o(1)) \log_2 d. \quad (4.4)$$

5 The Gaussian label gadget

We next realize the arbitrary finite-group translations from [Theorem 4.1](#) by adding one Euclidean label coordinate per binary translation.

For a spacing $a > 0$ and a residual $t \in [0, a)$, define the two normalized one-dimensional branch weights

$$h_0(t) := a\gamma_1(t), \quad h_1(t) := a\gamma_1(t - a). \quad (5.1)$$

Lemma 5.1 (A heavy label inside every dense label set). *Fix $\alpha \in (0, 1]$ and $a > 0$. There is a constant $C_{\alpha, a}$ such that the following holds for every $\ell \geq 1$. Let $\mathcal{A} \subseteq \{0, 1\}^\ell$ satisfy $|\mathcal{A}| \geq \alpha 2^\ell$, and let $t_1, \dots, t_\ell \in [0, a)$. Then there is $\varepsilon \in \mathcal{A}$ such that*

$$\prod_{i=1}^{\ell} h_{\varepsilon_i}(t_i) \geq e^{-C_{\alpha, a}\sqrt{\ell}} \prod_{i=1}^{\ell} \sqrt{h_0(t_i)h_1(t_i)}. \quad (5.2)$$

Consequently,

$$\prod_{i=1}^{\ell} h_{\varepsilon_i}(t_i) \geq e^{-C_{\alpha, a}\sqrt{\ell}} \left(\frac{a}{\sqrt{2\pi}} e^{-a^2/4} \right)^\ell. \quad (5.3)$$

The base in parentheses is maximized at $a = \sqrt{2}$, where it equals $(\pi e)^{-1/2}$.

Proof. Choose a uniform random $\varepsilon \in \{0, 1\}^\ell$ and put

$$X_i := \log h_{\varepsilon_i}(t_i).$$

Then

$$\mathbb{E}X_i = \frac{1}{2} \log(h_0(t_i)h_1(t_i)).$$

Moreover,

$$|\log h_0(t) - \log h_1(t)| = \frac{1}{2} |(a - t)^2 - t^2| \leq \frac{a^2}{2}.$$

Hoeffding's inequality [\[3\]](#) therefore gives

$$\mathbb{P} \left(\sum_i X_i < \sum_i \mathbb{E}X_i - C_{\alpha, a}\sqrt{\ell} \right) < \frac{\alpha}{2}$$

for a suitable constant $C_{\alpha, a}$. Since $\mathcal{A}$ has density at least α , it meets the complementary high-weight set, proving [\(5.2\)](#).

For [\(5.3\)](#), observe that

$$\sqrt{h_0(t)h_1(t)} = \frac{a}{\sqrt{2\pi}} \exp \left(-\frac{t^2 + (a - t)^2}{4} \right) \geq \frac{a}{\sqrt{2\pi}} e^{-a^2/4},$$

because $t^2 + (a - t)^2 \leq a^2$ on $[0, a]$. Finally, differentiating $\log a - a^2/4$ shows that the maximum occurs at $a = \sqrt{2}$, giving $(\pi e)^{-1/2}$. $\square$

The remainder of the note uses the optimal spacing

$$a := \sqrt{2}. \quad (5.4)$$

6 The extended vertical lattice

Let $C_0 \leq \mathbb{F}_Q^d$, $S = B + C_0$, and $w_1, \dots, w_\ell \in \mathbb{F}_Q^d$ be as above. For each i , let $\widehat{w}_i \in \{0, \dots, Q - 1\}^d$ be the standard integer representative and put

$$u_i := \frac{\widehat{w}_i}{Q} \in \mathbb{R}^d.$$

Set

$$D := d + \ell.$$

Construction 6.1 (Two-stage vertical lattice). *Define*

$$\widehat{\Lambda}_v := \left\{ \left(\tau^{-1/2} \left(\lambda + \sum_{i=1}^{\ell} n_i u_i \right), an \right) : \lambda \in \Lambda_0, n \in \mathbb{Z}^{\ell} \right\} \subseteq \mathbb{R}^d \oplus \mathbb{R}^{\ell}. \quad (6.1)$$

Then $\widehat{\Lambda}_v$ is a full-rank lattice and

$$\det \widehat{\Lambda}_v = \tau^{-d/2} Q^{-k} a^{\ell}. \quad (6.2)$$

Moreover,

$$\Psi(\mathbb{Z}^d) \subseteq \widehat{\Lambda}_v, \quad \Psi(\eta) := \left(\tau^{-1/2} \eta, 0 \right), \quad (6.3)$$

where $\Psi: \mathbb{R}^d \rightarrow \mathbb{R}^D$ is injective.

Proof. Choose a basis of Λ_0 . Relative to that basis and the standard basis of $\mathbb{Z}^{\ell}$, a generating matrix for (6.1) is block triangular, with diagonal blocks $\tau^{-1/2} B_0$ and $a I_{\ell}$, where B_0 is a basis matrix for Λ_0 . This proves discreteness, full rank, and (6.2). Since $\mathbb{Z}^d \subseteq \Lambda_0$, (6.3) follows. $\square$

The following theorem is the new vertical estimate.

Theorem 6.2 (Two-stage vertical weight). *There is a nonnegative deterministic sequence $\xi_d \rightarrow 0$ such that, for every sufficiently large d , the lattice in Theorem 6.1 has the following property. For every $y = (y^{\text{old}}, y^{\text{lab}}) \in \mathbb{R}^d \oplus \mathbb{R}^{\ell}$, there are a lattice point $\lambda_y \in \widehat{\Lambda}_v$ and a point $\tilde{t}_y \in [0, 1]^d$ such that the 2^d vertical residuals*

$$y - \lambda_y - \Psi(\eta), \quad \eta \in \{0, 1\}^d,$$

have total normalized Gaussian weight

$$W_y := \det \widehat{\Lambda}_v \sum_{\eta \in \{0, 1\}^d} \gamma_D(y - \lambda_y - \Psi(\eta)) \quad (6.4)$$

obey the uniform two-sided estimate

$$d^{-a_* - \xi_d} \leq W_y \leq d^{-1/2 + \xi_d}, \quad (6.5)$$

where

$$a_* := \frac{1}{2} \log_2(2\pi e). \quad (6.6)$$

The normalized weights on $\{0, 1\}^d$ are exactly $P_{\tilde{t}_y}$. Consequently,

$$H_{\infty}(P_{\tilde{t}_y}) \geq ds_{\infty} - C \sqrt{ds_{\infty} \log d} - C \Delta_d, \quad (6.7)$$

$$H(P_{\tilde{t}_y}) \leq ds + C \sqrt{ds \log d} + C \Delta_d. \quad (6.8)$$

All residuals satisfy

$$\|y - \lambda_y - \Psi(\eta)\|_2^2 \leq \frac{d}{\tau} + a^2 \ell = 2d \log d + O(\log d). \quad (6.9)$$

Finally,

$$D = d + O(\log d) = (1 + o(1))d. \quad (6.10)$$

Proof. Fix $y = (y^{\text{old}}, y^{\text{lab}})$. For each label coordinate, choose $n_i^0 \in \mathbb{Z}$ so that

$$t_i := y_i^{\text{lab}} - a n_i^0 \in [0, a).$$

Put

$$w := \sqrt{\tau} y^{\text{old}} - \sum_{i=1}^{\ell} n_i^0 u_i.$$

Choose $z_0 \in \mathbb{Z}^d$ such that $w_0 := w - z_0 \in [0, 1]^d$, and let $z \in \mathbb{F}_Q^d$ be the label of the grid cell containing w_0 .

By [Theorem 4.1](#), the set

$$\mathcal{A}_z := \left\{ \varepsilon \in \{0, 1\}^\ell : z - \sum_i \varepsilon_i w_i \in S \right\}$$

has size at least $\alpha_{\sigma_0} 2^\ell$. Apply [Theorem 5.1](#) to $\mathcal{A}_z$ and the residuals $t_1, \dots, t_\ell$. We obtain an $\varepsilon \in \mathcal{A}_z$ such that

$$\prod_{i=1}^{\ell} a \gamma_1(t_i - a \varepsilon_i) \geq e^{-C\sqrt{\ell}} (\pi e)^{-\ell/2}. \quad (6.11)$$

Since $z - \sum_i \varepsilon_i w_i \in S = B + C_0$, write

$$z - \sum_i \varepsilon_i w_i = b + c$$

with $b \in B$ and $c \in C_0$. Let $\widehat{z}, \widehat{b}, \widehat{c}, \widehat{w}_i$ be the standard integer representatives. There is a $\kappa \in \mathbb{Z}^d$ such that

$$\widehat{z} - \sum_i \varepsilon_i \widehat{w}_i = \widehat{b} + \widehat{c} - Q\kappa.$$

Writing $w_0 = \widehat{z}/Q + u$ with $u \in [0, Q^{-1})^d$, define

$$\widetilde{t}_y := w_0 + \kappa - \frac{\widehat{c}}{Q} - \sum_i \varepsilon_i \frac{\widehat{w}_i}{Q} = \frac{\widehat{b}}{Q} + u \in \mathcal{Q}_b.$$

Set

$$\lambda^{\text{old}} := z_0 - \kappa + \frac{\widehat{c}}{Q} \in \Lambda_0$$

and

$$\lambda_y := \left(\tau^{-1/2} \left(\lambda^{\text{old}} + \sum_i (n_i^0 + \varepsilon_i) u_i \right), a(n^0 + \varepsilon) \right) \in \widehat{\Lambda}_v.$$

Then, for every $\eta \in \{0, 1\}^d$,

$$y - \lambda_y - \Psi(\eta) = \left(\frac{\widetilde{t}_y - \eta}{\sqrt{\tau}}, t - a\varepsilon \right). \quad (6.12)$$

Using [\(6.2\)](#), [\(6.12\)](#), and [Theorem 2.2](#), we obtain the exact product formula

$$W_y = Q^{-k} P_\tau^d f^{\otimes d}(\widetilde{t}_y) \prod_{i=1}^{\ell} a \gamma_1(t_i - a \varepsilon_i). \quad (6.13)$$

Because $\widetilde{t}_y$ and t_b lie in the same grid cell, [\(2.9\)](#), [\(3.2\)](#), and [\(2.5\)](#) give a nonnegative deterministic sequence $\delta_d \rightarrow 0$ such that, uniformly in y ,

$$d^{-1/2-\delta_d} \leq \beta f^{\otimes d}(\widetilde{t}_y) \leq d^{-1/2+\delta_d}. \quad (6.14)$$

Together with [\(3.1\)](#) and [\(2.8\)](#), after enlarging δ_d if necessary, this yields

$$d^{-1/2-\delta_d} \leq Q^{-k} P_\tau^d f^{\otimes d}(\widetilde{t}_y) \leq d^{-1/2+\delta_d}. \quad (6.15)$$

Put $b_* := \frac{1}{2} \log_2(\pi e)$, so that $a_* = \frac{1}{2} + b_*$. For suitable absolute constants $C_0, C_1 > 0$, [\(6.11\)](#), [\(6.15\)](#), and the upper bound in [\(4.4\)](#) give

$$W_y \geq d^{-1/2-\delta_d} M^{-b_*} e^{-C_1 \sqrt{\ell}} \geq d^{-a_* - \xi_d},$$

where we set

$$\xi_d := \delta_d + \frac{b_* \log(C_0 \log Q) + C_1 \sqrt{\ell}}{\log d}. \quad (6.16)$$

This is a nonnegative deterministic sequence tending to zero, by (2.1) and (4.4). For the upper bound, the choice $a = \sqrt{2}$ gives

$$a\gamma_1(t_i - a\varepsilon_i) \leq \frac{a}{\sqrt{2\pi}} = \pi^{-1/2} < 1$$

in every label coordinate. Combining this with the upper bound in (6.15), and noting that $\xi_d \geq \delta_d$, proves the upper bound in (6.5) for the same sequence (6.16).

The label factor in (6.13) is independent of η . Hence the normalized law on the d -dimensional Boolean cube is exactly P_{t_y} . The entropy estimates follow from (2.6), (2.7), and (2.10). Finally, every old coordinate of $\tilde{t}_y - \eta$ lies in $[-1, 1]$, while every label residual lies in $[-a, a]$. This proves (6.9), and (6.10) follows from (4.4). $\square$

7 Partial-rank finite-index lifting

The weighted lifting theorem in the companion paper [5, Theorem 4.4] assumes that the Boolean directions form a finite-index full-rank sublattice of the vertical lattice. Our vertical space has dimension $D = d + \ell$, while the Boolean directions have rank d . We first recall the pattern language used by weighted amplification and then prove the required rectangular-matrix extension.

For a positive integer N , write $[N] := \{1, \dots, N\}$. If $\Lambda \subseteq \mathbb{R}^m$ is a full-rank lattice and $J \subseteq \mathbb{R}^m$ is Borel, define its *uncovered proportion* by

$$\bar{\rho}(\Lambda + J) := 1 - \mu_\Lambda(\pi_\Lambda(J)),$$

where $\pi_\Lambda: \mathbb{R}^m \rightarrow \mathbb{R}^m/\Lambda$ is the quotient map and μ_Λ is Haar probability measure.

For $I \subseteq [d]$, a *labeled pattern* is a pair (P, α) consisting of a nonempty set $P \subseteq \{0, 1\}^I$ and a map $\alpha: P \rightarrow [N]$. Fixing a coordinate $j \in I$ to a value $e \in \{0, 1\}$ gives an e -slice on $I \setminus \{j\}$; an empty slice is discarded, and a nonempty slice retains the labels of its surviving vertices. A family of triples (I, P, α) is *slice-closed* if it contains every nonempty coordinate slice of each of its members. For a family $\mathcal{A}$ of labeled patterns on $[d]$, let $\text{scl}(\mathcal{A})$ be the smallest slice-closed family containing $([d], P, \alpha)$ for every $(P, \alpha) \in \mathcal{A}$.

We use the following weighted amplification theorem from the companion paper [5, Theorem 4.3]. Let $\mathbb{T}$ be a compact abelian group with Haar probability measure μ , let $U_1, \dots, U_N \subseteq \mathbb{T}$ be measurable with $\mu(U_j) \leq \delta_j$, and let $\mathcal{C}$ be a finite slice-closed family. If $L \geq 2|\mathcal{C}|$, then there are $z_1, \dots, z_d \in \mathbb{T}$ such that, for every $(I, P, \alpha) \in \mathcal{C}$,

$$\mu \left(\bigcap_{\omega \in P} \left(U_{\alpha(\omega)} - \sum_{i \in I} \omega_i z_i \right) \right) \leq L^{|P|-1} \prod_{\omega \in P} \delta_{\alpha(\omega)}. \quad (7.1)$$

Theorem 7.1 (Partial-rank weighted lifting). *Let $\mathbb{R}^n = E \oplus F$ be an orthogonal decomposition, where $\dim E = m$ and $\dim F = D$. Let $K \subseteq E \oplus F$ be compact and define $K_y := \{x \in E: x + y \in K\}$. Suppose that:*

- (i) $J_1, \dots, J_N \subseteq E$ are compact, $\Lambda_h \subseteq E$ is a lattice, and $\delta_j \in [0, 1]$ satisfy

$$\bar{\rho}(\Lambda_h + J_j) \leq \delta_j \quad \text{for every } j \in [N];$$

- (ii) $\Lambda_v \subseteq F$ is a lattice and $\Psi: \mathbb{R}^d \rightarrow F$ is an injective linear map satisfying $\Psi(\mathbb{Z}^d) \subseteq \Lambda_v$;

- (iii) $\mathcal{A}$ is a finite family of labeled patterns (P, α) with $\emptyset \neq P \subseteq \{0, 1\}^d$ and $\alpha: P \rightarrow [N]$, such that for every $w \in F$ there are $\lambda \in \Lambda_v$ and $(P, \alpha) \in \mathcal{A}$ with

$$J_{\alpha(\omega)} \subseteq K_{w-\lambda-\Psi(\omega)} \quad \text{for every } \omega \in P.$$

Let $\mathcal{C} := \text{scl}(\mathcal{A})$ and $L := 2|\mathcal{C}|$. Then there is a lattice $\Lambda \subseteq E \oplus F$ with

$$\det \Lambda = \det \Lambda_h \det \Lambda_v$$

and

$$\bar{\rho}(\Lambda + K) \leq \max_{(P, \alpha) \in \mathcal{A}} L^{|P|-1} \prod_{\omega \in P} \delta_{\alpha(\omega)}.$$

Proof. Choose a linear isomorphism $S: E \rightarrow \mathbb{R}^m$ satisfying $S(\Lambda_h) = \mathbb{Z}^m$. Apply $S \oplus \text{id}_F$ to K and the J_j , retaining the same notation for their images. Section inclusions and uncovered proportions are unchanged.

Choose a basis map $B: \mathbb{R}^D \rightarrow F$ satisfying $B(\mathbb{Z}^D) = \Lambda_v$. Since $\Psi(\mathbb{Z}^d) \subseteq \Lambda_v$, there is an integer matrix

$$M \in M_{D \times d}(\mathbb{Z})$$

of rank d such that $\Psi = B \circ M$.

Let $\mathbb{T}^m := \mathbb{R}^m / \mathbb{Z}^m$, let $\pi: \mathbb{R}^m \rightarrow \mathbb{T}^m$ be the quotient map, and put $U_j := \mathbb{T}^m \setminus \pi(J_j)$. Then U_j is Borel and has Haar measure at most δ_j . Apply (7.1) to $\mathcal{C}$, obtaining shifts $z_1, \dots, z_d \in \mathbb{T}^m$. Write $z := (z_1, \dots, z_d)$.

Consider the torus homomorphism

$$(\mathbb{T}^m)^D \longrightarrow (\mathbb{T}^m)^d, \quad \chi \longmapsto M^\top \chi,$$

which is onto. Indeed, Smith normal form for the full-column-rank integer matrix M reduces the claim to surjectivity of multiplication by a nonzero integer on $\mathbb{T}^m$. Choose $\chi = (\chi_1, \dots, \chi_D)$ with $M^\top \chi = z$, and choose representatives $\hat{\chi}_j \in \mathbb{R}^m$.

For $\nu \in \mathbb{Z}^D$, write $\nu \cdot \hat{\chi} := \sum_{j=1}^D \nu_j \hat{\chi}_j$ and define the shear lattice

$$\Gamma := \{(k - \nu \cdot \hat{\chi}, B\nu) : k \in \mathbb{Z}^m, \nu \in \mathbb{Z}^D\} \subseteq \mathbb{R}^m \oplus F.$$

Its generating matrix is block triangular with diagonal blocks I_m and B , so $\det \Gamma = \det \Lambda_v$.

Let $\mathcal{D}_h := [0, 1]^m$, and let $\mathcal{D}_v$ be a half-open fundamental domain for Λ_v . Reduction first in the vertical coordinate and then in the horizontal coordinate shows that $\mathcal{D}_h \times \mathcal{D}_v$ is a fundamental domain for Γ , up to a null boundary. Set

$$\mathcal{H} := (\mathcal{D}_h \times \mathcal{D}_v) \setminus (\Gamma + K), \quad \mathcal{H}_w := \{x \in \mathcal{D}_h : (x, w) \in \mathcal{H}\}.$$

Because K is compact and Γ is discrete, $\Gamma + K$ is closed, so these hole sets are measurable.

Fix $w \in \mathcal{D}_v$ and choose a witness $\lambda = B\nu_0$ with pattern (P, α) . For $\omega \in P$, put $\nu_\omega := \nu_0 + M\omega$. The points of Γ with vertical coordinate $\lambda + \Psi(\omega) = B\nu_\omega$ are

$$(k - \nu_\omega \cdot \hat{\chi}, B\nu_\omega), \quad k \in \mathbb{Z}^m.$$

If $x \in \mathcal{H}_w$, then

$$\pi(x + \nu_\omega \cdot \hat{\chi}) \in U_{\alpha(\omega)} \quad \text{for every } \omega \in P.$$

Otherwise, for some $k \in \mathbb{Z}^m$ the horizontal difference $x - k + \nu_\omega \cdot \hat{\chi}$ would lie in $J_{\alpha(\omega)}$; the witness inclusion would then put the difference between (x, w) and the corresponding point of Γ inside K , contradicting $x \in \mathcal{H}_w$.

In $\mathbb{T}^m$, the phase decomposes as

$$(\nu_0 + M\omega) \cdot \chi = \nu_0 \cdot \chi + \omega \cdot (M^\top \chi) = \nu_0 \cdot \chi + \omega \cdot z.$$

Therefore the image of $\mathcal{H}_w$ under the measure-preserving map $x \mapsto \pi(x + \nu_0 \cdot \hat{\chi})$ is contained in

$$\bigcap_{\omega \in P} (U_{\alpha(\omega)} - \omega \cdot z).$$

Equation (7.1) gives

$$\text{vol}_m(\mathcal{H}_w) \leq L^{|P|-1} \prod_{\omega \in P} \delta_{\alpha(\omega)}.$$

This bound is uniform in w , even though no measurable choice of witnesses is required. Tonelli's theorem now yields

$$\bar{\rho}(\Gamma + K) = \frac{1}{\det \Lambda_v} \int_{\mathcal{D}_v} \text{vol}_m(\mathcal{H}_w) \, dw \leq \max_{(P, \alpha) \in \mathcal{A}} L^{|P|-1} \prod_{\omega \in P} \delta_{\alpha(\omega)}.$$

Finally, apply $S^{-1} \oplus \text{id}_F$. This transformation preserves the uncovered proportion and changes the covolume from $\det \Lambda_v$ to $\det \Lambda_h \det \Lambda_v$, proving both conclusions. $\square$

8 Proof of the covering bound

We now combine the new vertical lattice with the section discretization, common-lattice estimate, entropy pruning, and completion steps of the companion paper. The details below are included to distinguish the number of section labels from the larger slice-closed pattern family and to verify that the added label coordinates preserve every required asymptotic range.

Proof of Theorem 1.1. Put

$$\ell_n := \log \log n.$$

For a sufficiently large absolute constant A , choose

$$d := \left\lceil A \ell_n^{4/3} (\log \ell_n)^{2/3} \right\rceil.$$

By (2.3), increasing A if necessary gives

$$ds_\infty \geq 8\ell_n, \quad ds = O(\ell_n). \quad (8.1)$$

Moreover,

$$d = \ell_n^{4/3} (\log \ell_n)^{2/3+o(1)} = \ell_n^{4/3+o(1)}. \quad (8.2)$$

Construct the vertical lattice in Theorem 6.1 and use the notation of Theorem 6.2. Thus $D = d + O(\log d) = (1 + o(1))d$. Put

$$R := \sqrt{\frac{d}{\tau} + a^2 \ell} = \sqrt{2d \log d + O(\log d)}.$$

Let $c_1, c_3, c_4 > 0$ be the constants in Proposition 4.1 of the companion paper. Because d is a fixed power of $\log \log n$, one has

$$D = o(n), \quad D \log n = o(n), \quad 4^d D = o(n), \quad D \leq n^{c_1}, \quad 2DR < n^{c_4} \quad (8.3)$$

for all sufficiently large n .

Fix an arbitrary n -dimensional convex body K and put $m := n - D$. Fix an absolute constant C_{sec} at least as large as the label-count constant in Proposition 4.1 of the companion paper and such that

$$C_{\text{sec}} \geq 2(1 + c_{\text{RS}}). \quad (8.4)$$

Apply Proposition 4.1 of the companion paper [5], with vertical dimension D and radius R . After an affine normalization, it gives an orthogonal decomposition $\mathbb{R}^n = E \oplus F$, with $\dim E = m$ and $\dim F = D$, and convex bodies $J_1, \dots, J_N \subseteq E$ such that

$$N \leq N_0 := (C_{\text{sec}} m^2)^D. \quad (8.5)$$

For every $v \in F$ with $\|v\|_2 \leq R$, at least one label $j \in [N]$ satisfies $J_j \subseteq K_v$ and

$$\text{vol}_m(J_j) = (1 + O(m^{-1} + n^{-c_3})) \text{vol}_n(K) \gamma_D(v), \quad (8.6)$$

where the error is deterministic and uniform in K and v .

Identify the vertical space of Theorem 6.1 isometrically with F . Write $D_v := \det \hat{\Lambda}_v$. For each $y \in F$, choose any witness λ_y supplied by Theorem 6.2, and define

$$\mathbf{w}_{y,\eta} := D_v \gamma_D(y - \lambda_y - \Psi(\eta)), \quad q_{y,\eta} := \frac{\mathbf{w}_{y,\eta}}{\mathbf{W}_y}.$$

All residuals lie in the ball of radius R . By Theorem 6.2, there is a deterministic sequence $\kappa_d \rightarrow 0$ such that, uniformly in y ,

$$d^{-a_* - \xi_d} \leq \mathbf{W}_y \leq d^{-1/2 + \xi_d}, \quad (8.7)$$

$$-\log \max_{\eta} q_{y,\eta} \geq (1 - \kappa_d) ds_\infty, \quad (8.8)$$

$$H(q_y) \leq (1 + \kappa_d) ds. \quad (8.9)$$

Fix a sufficiently large absolute constant $B_0 \geq 8$ and define the target density by the exact formula

$$\rho := B_0 n \log n d^{a_* + \xi_d}. \quad (8.10)$$

Choose the horizontal covolume

$$D_h := \frac{\text{vol}_n(K)}{\rho D_v}.$$

The ideal load at a Boolean vertex is

$$\hat{u}_{y,\eta} := \rho \mathbf{W}_{y,\eta}.$$

The lower bound in (8.7) gives

$$\sum_{\eta} \hat{u}_{y,\eta} = \rho \mathbf{W}_y \geq B_0 n \log n. \quad (8.11)$$

We also need an upper bound on every individual load. Equations (8.7) and (8.8) give

$$\max_{\eta} \hat{u}_{y,\eta} \leq B_0 n \log n d^{a_* - 1/2 + 2\xi_d} e^{-(1-\kappa_d)ds_{\infty}} = o(n),$$

where (8.1) absorbs the fixed power of d . Set

$$b_0 := \frac{1}{4} \log \frac{27}{16}, \quad \eta_m := b_0 m - 3 \log m.$$

It follows that

$$\max_{\eta} \hat{u}_{y,\eta} \leq \frac{1}{4} \eta_m \quad (8.12)$$

uniformly in y , for all sufficiently large n .

For each residual $y - \lambda_y - \Psi(\eta)$, choose a section label $\alpha_y(\eta)$ satisfying (8.6), and put $u_j := \text{vol}_m(J_j)/D_h$. The label map need not be injective; repeated labels are counted with their Boolean-vertex multiplicities in all sums and products below. There is a nonnegative deterministic sequence $\zeta_n \rightarrow 0$, independent of K, y, η , with $\zeta_n \leq 1/2$ for all sufficiently large n , such that

$$(1 - \zeta_n) \hat{u}_{y,\eta} \leq u_{\alpha_y(\eta)} \leq (1 + \zeta_n) \hat{u}_{y,\eta}. \quad (8.13)$$

Set

$$T_0 := 4 \log(C_{\text{sec}} N_0).$$

Then $T_0 = O(D \log n) = o(n \log n)$. For each y , retain the pattern

$$P_y := \{\eta \in \{0, 1\}^d : \hat{u}_{y,\eta} \geq 2T_0\}.$$

Every retained vertex has actual load at least T_0 and, by (8.13) and (8.12), at most η_m .

For completeness, the entropy-pruning estimate can be seen directly. If $\eta \notin P_y$, then

$$q_{y,\eta} < \frac{2T_0}{\rho \mathbf{W}_y}.$$

For any probability vector q and $0 < \theta < 1$,

$$\sum_{q_x \leq \theta} q_x \leq \frac{H(q)}{\log(1/\theta)}.$$

Here (8.11), (8.9), and $T_0 = O(D \log n)$ imply

$$\frac{H(q_y)}{\log(\rho \mathbf{W}_y / (2T_0))} = O\left(\frac{\ell_n}{\log n}\right) = o(1).$$

Combining this estimate with (8.13) gives the uniform retained-load bound

$$\sum_{\eta \in P_y} u_{\alpha_y(\eta)} \geq (1 - o(1)) B_0 n \log n. \quad (8.14)$$

In particular, every P_y is nonempty for large n .

Let $\mathcal{A}_0$ be the family of all labeled retained patterns $(P_y, \alpha_y|_{P_y})$ that can arise from a vertical translate, an admissible vertical witness, and admissible section labels. This is an existential family; no measurable global choice of witnesses is needed. It is finite because P_y is a subset of the finite Boolean cube and every label lies in $[N]$.

Let $\mathcal{I}_{\text{act}} \subseteq [N]$ be the set of labels occurring in members of $\mathcal{A}_0$, and put $N_{\text{act}} := |\mathcal{I}_{\text{act}}| \leq N_0$. Discard the unused bodies and reindex the bodies in $\mathcal{I}_{\text{act}}$, together with every label map in $\mathcal{A}_0$, by $[N_{\text{act}}]$. Denote the resulting family of labeled patterns by $\mathcal{A}$. By (8.4),

$$T_0 = 4 \log(C_{\text{sec}} N_0) \geq 2 \log(2(1 + c_{\text{RS}}) N_{\text{act}}).$$

Thus the precise load-window and union-bound hypotheses of Lemma 4.2 of the companion paper hold. That lemma gives a lattice $\Lambda_h \subseteq E$ of covolume D_h such that

$$\bar{\rho}(\Lambda_h + J_j) \leq e^{-u_j/2} \quad (8.15)$$

for every active label j .

Put $\mathcal{C} := \text{scl}(\mathcal{A})$ and $L := 2|\mathcal{C}|$. Every member of $\mathcal{C}$ is a labeled nonempty subpattern on some subset of the d Boolean coordinates, so

$$\log L \leq \log 2 + d \log 2 + 2^d \log(N_0 + 1). \quad (8.16)$$

The selected correction vector is common to all vertices of the d -dimensional Boolean cube. It may change a section label, but the section discretization already provides one alphabet of at most N_0 labels for every point in the D -dimensional ball. Thus no additional factor 2^ℓ appears in (8.16); only the replacement of d by D in N_0 is required. Since $|P| - 1 \leq 2^d$ and $\log(N_0 + 1) = O(D \log n)$, (8.3) gives

$$(|P| - 1) \log L = O(4^d D \log n) = o(n \log n) \quad (8.17)$$

uniformly over $(P, \alpha) \in \mathcal{A}$.

Apply Theorem 7.1 with $\delta_j = e^{-u_j/2}$. Equations (8.14) and (8.17) produce a lattice $\Lambda \subseteq \mathbb{R}^n$ with $\det \Lambda = D_h D_v$. With the convention $\log 0 = -\infty$, it satisfies

$$\log \bar{\rho}(\Lambda + K) \leq o(n \log n) - \frac{1}{2}(1 - o(1)) B_0 n \log n < -2n \log n.$$

Hence

$$\bar{\rho}(\Lambda + K) < n^{-2n} < (n^n + 1)^{-1}.$$

Rogers' completion lemma [9, Lemma 4] gives $\Lambda + (1 + 1/n)K = \mathbb{R}^n$. Equivalently, $\Lambda' := (1 + 1/n)^{-1}\Lambda$ satisfies $\Lambda' + K = \mathbb{R}^n$, and

$$\frac{\text{vol}_n(K)}{\det \Lambda'} = \left(1 + \frac{1}{n}\right)^n \frac{\text{vol}_n(K)}{D_h D_v} \leq e\rho.$$

Therefore

$$\theta_L(K) \leq e B_0 n \log n d^{a_* + \xi_d}. \quad (8.18)$$

Define

$$\alpha_* := \frac{4}{3} a_* = \frac{2}{3} \log_2(2\pi e)$$

and, for all sufficiently large n , put

$$\varepsilon_n := \max \left\{ 0, \frac{(a_* + \xi_d) \log d}{\log \ell_n} - \alpha_* \right\}.$$

Set $\varepsilon_n := 0$ for the remaining finitely many n . Equation (8.2) and $\xi_d \rightarrow 0$ imply that $\varepsilon_n \rightarrow 0$, independently of K , and

$$d^{a_* + \xi_d} \leq (\log \log n)^{\alpha_* + \varepsilon_n}.$$

Substituting this estimate into (8.18) proves Theorem 1.1 with $C = e B_0$. Affine invariance permits us to undo the normalization without changing the covering density. $\square$

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